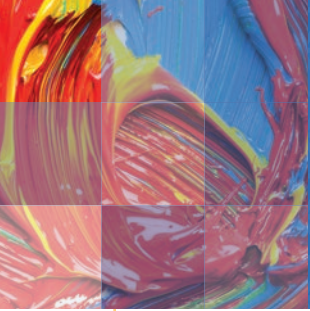




Linear Programming

Systems of linear inequalities are used in linear programming.

Definitions

Linear programming is the process of maximizing or minimizing a linear function subject to a set of conditions, called **constraints**, that are linear inequalities.

Each constraint in a linear programming problem is a linear inequality. The set of constraints that must be satisfied forms a system of linear inequalities. The graph of the solutions forms the **feasible region**, which is the set of all the ordered pairs that satisfy the constraints and are possible solutions to the problem.

► **Think About It** The word *feasible* means “possible to achieve.” Ordered pairs outside the feasible region do not satisfy all of the constraints, so they are not possible solutions to the problem.

Determining a Feasible Region

EXAMPLE 1

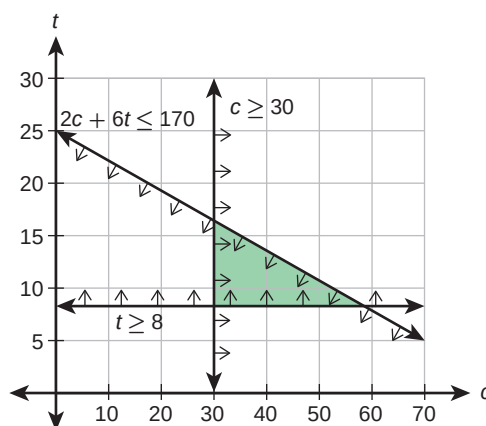
Graph the feasible region.

A
$$\begin{cases} c \geq 30 \\ t \geq 8 \\ 2c + 6t \leq 170 \end{cases}$$

SOLUTION

Graph each inequality.

Shade the region where all three inequalities intersect. The feasible region is bounded because the boundary lines form a closed figure.

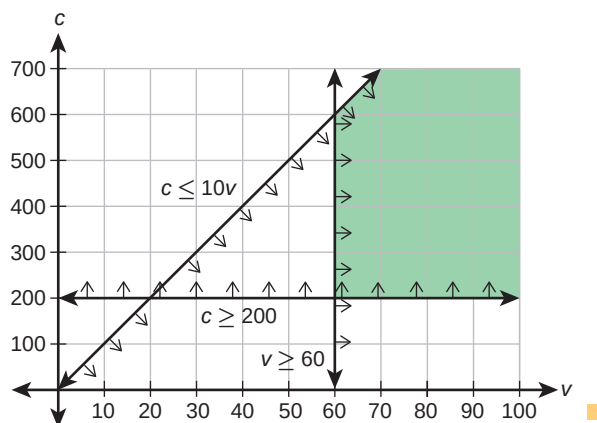


B
$$\begin{cases} v \geq 60 \\ c \geq 200 \\ c \leq 10v \end{cases}$$

SOLUTION

Graph each inequality.

Shade the region where all three inequalities intersect. The feasible region is unbounded, and there is an infinite number of solutions.



Finding an Optimal Solution

Once you know a feasible region, you can find the optimal, or best, solution for the problem. An **optimal solution** is a solution that maximizes or minimizes the objective.

Definition

An **objective function** is a linear function that models a quantity that is to be optimized in a linear programming problem.

An optimal solution to a linear programming problem will always occur at one of the corner points, called vertex points, of the feasible region. The maximum and minimum values of the objective function will always occur at vertex points.

EXAMPLE 2

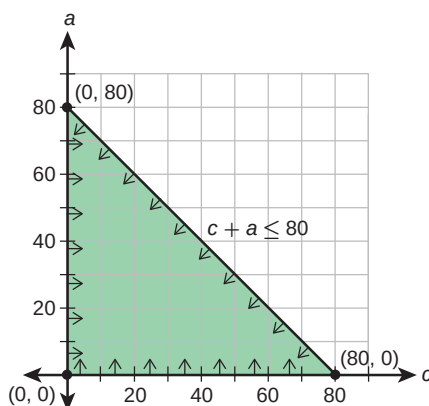
Consider the constraints.

$$\begin{cases} c \geq 0 \\ a \geq 0 \\ c + a \leq 80 \end{cases}$$

A Graph the feasible region.

SOLUTION

Determine the coordinates of the vertices of the feasible region. The vertices are at the intersections of the graphs of the constraints. The vertices are $(0, 0)$, $(80, 0)$, and $(0, 80)$.



B Find the maximum value of the objective function $P = 6c + 10a$.

SOLUTION

Evaluate the objective function at the vertices of the feasible region.

Vertex	$6c + 10a$	P
$(0, 0)$	$6 \cdot 0 + 10 \cdot 0$	0
$(80, 0)$	$6 \cdot 80 + 10 \cdot 0$	480
$(0, 80)$	$6 \cdot 0 + 10 \cdot 80$	800

The maximum value for P occurs at $c = 0$ and $a = 80$. The maximum is \$800. ■

Finding an Optimal Solution by Using a System of Equations

EXAMPLE 3

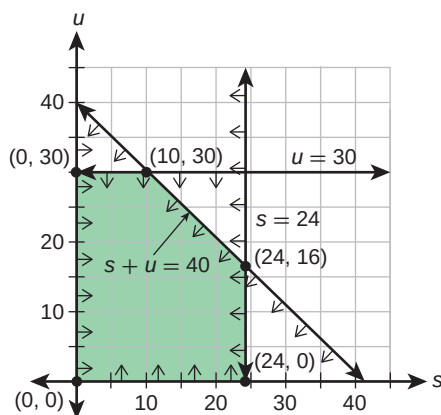
Find the maximum value of the objective function given the constraints.

$$\begin{cases} s \geq 0 \\ u \geq 0 \\ s \leq 24 \\ u \leq 30 \\ s + u \leq 40 \end{cases}$$

The objective function is $P = 12s + 8u$.

SOLUTION

Step 1 Graph the feasible region and determine the coordinates of the vertices. To find the coordinates of some intersections, you may need to solve a system of two equations.



Step 2 Evaluate the objective function at each vertex.

Vertex	$12s + 8u$	P
$(0, 0)$	$12 \cdot 0 + 8 \cdot 0$	0
$(0, 30)$	$12 \cdot 0 + 8 \cdot 30$	240
$(10, 30)$	$12 \cdot 10 + 8 \cdot 30$	360
$(24, 16)$	$12 \cdot 24 + 8 \cdot 16$	416
$(24, 0)$	$12 \cdot 24 + 8 \cdot 0$	288

Step 3 The value of P is greatest at $(24, 16)$, where the maximum value is 416. ■